Siyuan Lu

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Education

Zhejiang University / *Computer Science and Technology* / *Ph.D.* 2025 – 2030 (Expected)

Joint program with Westlake University and Shanghai Innovation Institute
Research: Agentic Systems and Agent Training Advisors: *Tao Lin*

Beijing University of Posts and Telecommunications / *Internet of Things* / *B.E.* 2021 – 2025

Joint program with Queen Mary University of London
Weighted Average Mark: 90.97/100 Rank: 7/182

Published Works

Don't Just Fine-tune the Agent, Tune the Environment

First Author ICLR 2026

Proposed Environment Tuning paradigm that enables agents to learn complex behaviors directly from problem instances without pre-collected expert trajectories. Through structured curriculum, environment augmentation, and fine-grained rewards, achieved superior tool-calling capabilities with only 400 samples.

Teach the Magnitude, Not the Direction: Verifier-Anchored Credit Assignment for Multi-Turn Multi-Step LLM Agents

Co-author EMNLP 2026 Main

Proposed CREST, a hierarchical credit-assignment framework for multi-turn, multi-step tool-use agents that combines turn-segmented verifiable rewards with entropy-gated self-teacher signals for token-level optimization. CREST preserves a verifier-bounded performance ceiling and consistently outperforms RL and distillation baselines on BFCL V3 and WildToolBench.

AWorld: Orchestrating the Training Recipe for Agentic AI

Co-author Technical Report

Proposed an open-source system for large-scale agent-environment interaction, achieving 14.6x speedup in experience collection through distributed deployment. The trained Qwen3-32B model reached 32.23% accuracy on GAIA test set, surpassing GPT-4o.

PreAct: Predicting Future in ReAct Enhances Agent's Planning Ability

Third Author COLING 2025

Proposed PreAct framework that integrates prediction, reasoning, and action. By leveraging predictive information, enables agents to provide broader and more strategic reasoning, outperforming ReAct methods on complex tasks.

Work Experience

GAIR 2026.3 – 2026.9

Visiting Student Generative Artificial Intelligence Research Lab (GAIR)

Mentor: *Pengfei Liu*
Participated in mid-training of the Shanghai Innovation Institute's self-developed large model; trained the Qwen3-30B-A3B base model to 76% on SWE-bench Verified through mid-training and SFT; maintained siirl-agentic (next-generation SIIRL adapted for agent training); researched asynchronous reinforcement learning algorithms for SWE tasks.

Ant Group 2025.6 – 2026.6

Research Intern AWorld Team, InclusionAI

Mentor: *Chenyi Zhuang*

Participated in research and development of the AWorld project, producing ICLR and EMNLP conference papers on post-training of large models in the tool-calling domain.

Open Source Contributions

T3RL 50+ Stars

Author Reinforcement Learning, Tool Calling

Built a multi-turn tool-calling reinforcement learning training framework on slime, reproducing and engineering the Environment Tuning paradigm: models learn through GRPO in real function execution environments instead of imitating static expert trajectories. Designed non-invasive training integration, multi-turn Gym environments, token-level trajectory alignment, and an asynchronous evaluation pipeline, supporting rapid adaptation of different models' inference and tool-calling formats via configuration, with two-stage curriculum training and evaluation completed on BFCL multi-turn tasks.

AWorld 1.2k Stars

Contributor Agentic AI, Self-Evolution, Multi-Agent Systems

AWorld is a self-evolving agent framework providing a complete agent learning loop (environment access, agent construction, experience retrieval, model training). The framework supports multi-agent collaboration, rich tool integration paradigms, advanced context management, and event-driven architecture, capable of capturing complete execution trajectories for continuous optimization. Main contributions: Participated in designing the training module, enabling training frameworks (such as verl) to leverage AWorld's agent abstraction and distributed environment deployment capabilities, accelerating rollout speed for large model reinforcement learning training and improving training efficiency.

Honors

First-Class Scholarship, Beijing University of Posts and Telecommunications 2023

Second-Class Scholarship, Beijing University of Posts and Telecommunications 2022

Skills

IELTS 7.0, proficient in English listening, speaking, reading, and writing

Familiar with Linux systems, proficient in common Linux commands, daily development in Linux environment

Familiar with Git, Docker, proficient in using GitHub for version control

Familiar with modern Python development, proficient in using latest Python toolchain such as uv, ruff

Familiar with sglang, vllm and other inference engine frameworks, as well as LLM reinforcement learning frameworks such as verl, slime, miles